

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF MATHEMATICS AND STATISTICS

TERM 2 2020

MATH3311 / MATH5335

**Mathematical Computing for Finance /
Computational Methods for Finance**

- (1) TIME ALLOWED – THREE (3) HOURS
- (2) TOTAL NUMBER OF QUESTIONS – 4
- (3) ANSWER ALL QUESTIONS
- (4) THE QUESTIONS ARE OF EQUAL VALUE
- (5) START EACH QUESTION ON A NEW PAGE.
- (6) THIS PAPER MAY BE RETAINED BY THE CANDIDATE

YOU ARE TO COMPLETE THE TEST UNDER STANDARD EXAM
CONDITIONS, WITH HANDWRITTEN SOLUTIONS.

YOU WILL THEN SUBMIT ONE OR MORE FILES CONTAINING YOUR
SOLUTIONS. **MAKE SURE YOU SUBMIT ALL YOUR ANSWERS.**

ONE OF THE SUBMITTED FILES MUST INCLUDE A PHOTOGRAPH OF
YOUR **STUDENT ID CARD** WITH THE **SIGNED, HANDWRITTEN**
STATEMENT:

“I declare that this submission is entirely my own original work.”

YOU CAN DELETE AND/OR RELOAD FILES UNTIL THE DEADLINE.

USE A SEPARATE PAGE CLEARLY MARKED QUESTION 1

1. i) Consider the floating point representation

$$(-1)^s(1.d_1d_2)_2 \times 2^e,$$

where $s, d_1, d_2, e \in \{0, 1\}$.

- Write down all the numbers which can be represented using this floating point representation.
 - How many bits need to be stored to save these floating point numbers?
 - What is the realmax in this case?
 - What is the realmin in this case?
 - What is the machine epsilon in this case?
- ii) Let $f(x) = \frac{1-\cos(x)}{x^2}$. The following Matlab code

```
x = linspace(-2e-8, 2e-8, 101);
plot(x, (1-cos(x))./x.^2)
```

produces Figure 1.

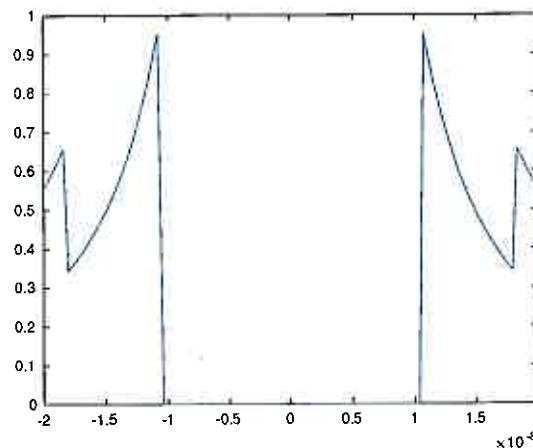


Figure 1: Matlab output

Recall that the Taylor series of $\cos$ is given by

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} \pm \dots$$

- Explain the behaviour of the Matlab output for $x \in [-10^{-8}, 10^{-8}] \setminus \{0\}$.
- How can you compute more accurate values of the function f for $x \in [-2 \times 10^{-8}, 2 \times 10^{-8}] \setminus \{0\}$?

Please see over ...

- c) What is the best approximation of $f(10^{-8})$ in double precision floating point arithmetic?
- iii) Let $n = 10^3$ and consider the $n \times n$ symmetric, strictly positive definite matrix A with real numbers as entries.

Computing the Cholesky factorization of A requires $n^3/3 + Cn^2$ floating point operations (for some constant $C > 0$).

- a) Assuming that each entry is stored in double precision, how much memory is required to store the matrix A ?
- b) Estimate how many flops are required to compute the Cholesky factorization of A .
- c) Estimate how long it will take to compute the Cholesky factorization of A on a 4 GHz quad core PC which can do 1 flop per clock cycle.

USE A SEPARATE PAGE CLEARLY MARKED QUESTION 2

2. i) Consider the quadratic spline $s : [-1, 1] \rightarrow \mathbb{R}$ given by $s(t) = s_0(t) = a_0 + b_0t + c_0t^2$ for $-1 \leq t \leq 0$ and $s(t) = s_1(t) = a_1 + b_1t + c_1t^2$ for $0 < t \leq 1$. The quadratic spline satisfies the conditions that

$$s_0(0) = s_1(0) \quad \text{and} \quad s'_0(0) = s'_1(0).$$

Further assume that $s'(-1) = 0$.

Find the coefficients $a_0, b_0, c_0, a_1, b_1, c_1$ of the quadratic spline which goes through the data points $(-1, 1), (0, 2), (1, 0)$.

- ii) Let $f : [0, 1] \rightarrow \mathbb{R}$ be given by

$$f(x) = A_0 + \sum_{k=1}^2 (A_k \cos(2\pi kx) + B_k \sin(2\pi kx)),$$

such that the function interpolates the data points

$$(0, 1), (1/5, 0), (2/5, 2), (3/5, 2), (4/5, 0),$$

i.e. $f(0) = 1, f(1/5) = 0, f(2/5) = 2, f(3/5) = 2, f(4/5) = 0$. Find the value of $\int_0^1 f(x) dx$.

- iii) Let N be a natural number and let $h = 2/N$. The composite trapezoidal rule with equal width h for a function $f : [-1, 1] \rightarrow \mathbb{R}$ is given by

$$Q_N(f) = h \left(\frac{1}{2} f(-1) + f(-1+h) + f(-1+2h) + \cdots + f(1-h) + \frac{1}{2} f(1) \right),$$

where $h = 2/N$.

Let $p_k(x) = c_0 + c_1x + \cdots + c_kx^k$ be a polynomial of degree k .

- a) Show that for $k = 0$ and $k = 1$ we have $\int_{-1}^1 p_k(x) dx = Q_N(p_k)$ for all natural numbers $N \in \{1, 2, 3, \dots\}$.

USE A SEPARATE PAGE CLEARLY MARKED QUESTION 3

3. i) Let X be a random variable with probability density function

$$p : [0, 3] \rightarrow \mathbb{R}$$

$$p(x) = cx^2$$

where $c > 0$ is a constant. Let $f : [0, 3] \rightarrow \mathbb{R}$ be given.

- a) Find the value of the constant c .
 - b) Write down the integral for the expectation value $\mathbb{E}[f(X)]$.
 - c) Find the inverse cumulative distribution function corresponding to p .
 - d) Let $Y \sim U([0, 1])$ be a uniformly distributed random variable. How can you use this random variable to generate samples from the random variable X ?
 - e) Write a Matlab code which generates 10 samples with pdf p given above.
- ii) Consider a European put option on an underlying stock with initial stock price $S_0 = 30$. Assume that the stock price either increases by a factor $u = 6/5$ or decreases by a factor $d = 5/6$, the time to expiry $T = 1/4$, the strike price $K = 30.5$, and the interest $r = 0.05$. The interest compounds continuously.
- Consider one time step of the binomial option pricing model, where the option price either increases to uS_0 or decreases to dS_0 at expiry time T . Consider a portfolio of Δ shares of the stock (long position) of the underlying asset and one European put option on the stock (long position).
- a) What is the value of Δ such that the value of the portfolio at expiry time is the same irrespective of whether the stock price increases to uS_0 or decreases to dS_0 ?
 - b) What is the value of the portfolio at time 0?
 - c) What is the value of the European put option at time 0?
- iii) Let $\alpha, \beta > 0$, $T > 0$ and X_0 be given constants. Consider the random walk

$$X_n = X_{n-1} + \alpha\delta + \beta\sqrt{\delta}Z_n, \quad n = 1, 2, \dots, N,$$

where $Z_1, Z_2, \dots, Z_N \sim \mathcal{N}(0, 1)$ are independent, standard normal random variables and $\delta = \frac{T}{N}$.

- a) Write a Matlab code which generates a random vector $(X_1, X_2, \dots, X_N)$.
- b) What is the distribution of X_N ?
- c) Find the mean and variance of X_N .

USE A SEPARATE PAGE CLEARLY MARKED QUESTION 4

4. The value $W(R, t)$ of an option on an underlying asset with price R at time t satisfies the Black–Scholes PDE

$$\frac{\partial W}{\partial t} + rR \frac{\partial W}{\partial R} + \frac{1}{2}\sigma^2 R^2 \frac{\partial^2 W}{\partial R^2} = rW \quad (1)$$

for $0 < R < \infty$ and $0 < t < T$. We assume that the prevailing risk-free interest rate r and the volatility σ are known positive constants.

Consider using (1) to price a put option, which gives the right to sell the underlying asset for the strike price L at expiry T .

- i) Carefully explain what the “initial” conditions are for this problem.
- ii) Carefully explain what the boundary conditions are for this problem.
- iii) The change of variables to log-moneyness s and time to expiry τ is

$$s = \log(R/L) \quad \text{and} \quad \tau = T - t.$$

This gives $V(s, \tau) = W(R, t) = W(Le^s, T - \tau)$.

- a) Find expressions for

$$\frac{\partial V}{\partial \tau}, \quad \frac{\partial V}{\partial s}, \quad \frac{\partial^2 V}{\partial s^2},$$

in terms of the partial derivatives of $W(R, t)$.

- b) Show that the Black and Scholes PDE (1) is equivalent to

$$-\frac{\partial V}{\partial \tau} + \left(r - \frac{\sigma^2}{2}\right) \frac{\partial V}{\partial s} + \frac{\sigma^2}{2} \frac{\partial^2 V}{\partial s^2} = rV \quad (2)$$

in terms of the log-moneyness s and the time to expiry τ .

- c) What are the initial conditions for the transformed problem?
- d) What are the boundary conditions for the transformed problem?
- iv) Let $s_{\max} > 0 > s_{\min}$ and consider the domain $s \in [s_{\min}, s_{\max}]$ and $\tau \in [0, T]$. Set up a grid $(s_n, \tau_j) = (s_{\min} + n\Delta s, j\Delta\tau)$ for $n = 0, 1, 2, \dots, M+1$ and $j = 0, 1, 2, \dots, K$, where

$$\Delta s = \frac{s_{\max} - s_{\min}}{M+1} \quad \text{and} \quad \Delta\tau = \frac{T}{K}.$$

Let $V_n^j \approx V(s_n, \tau_j)$ denote the approximate solution. At the time point τ_j and log-moneyness s_n give

- a) central difference approximations of $O(\Delta s^2)$ for the partial derivatives

$$\frac{\partial V}{\partial s} \quad \text{and} \quad \frac{\partial^2 V}{\partial s^2}.$$

Please see over ...

- b) the forward difference approximation of $O(\Delta\tau)$ for the partial derivative $\frac{\partial V}{\partial \tau}$.
- c) We want to use a finite difference method for computing an approximate solution V_n^j to equation (2).
- A) Show that the finite difference equation has the form

$$V_n^{j+1} = \alpha V_{n-1}^j + \beta V_n^j + \gamma V_{n+1}^j, \quad (3)$$

and find the coefficients α , β and γ .

- B) Write down the 'initial' and boundary conditions for V_n^j .

END OF EXAMINATION

3) i) a) ① $c = \frac{1}{9}$

b) ① $E(f(X)) = \int_0^3 f(x) p(x) dx = \frac{1}{9} \int_0^3 f(x) x^2 dx$

c) ② $p^{-1}(v) = 3 v^{1/3}$

d) ① $Y \sim U(0,1); X = p^{-1}(Y)$

e) ② $X = \text{rand}(10,1); X = 3 * X.^{(1/3)}$

ii) a) ③ $P_u = (K - uS_0)_+^0; P_d = (K - dS_0)_+^{5.5} \Delta S_0 u + (K - uS_0)_+ = \Delta S_0 d + (K - dS_0)_+$
 $\Delta = \frac{1}{2};$

b) ② $\Delta S_0 d + (K - dS_0)_+ = 18; \text{ at } T=0: 18e^{-rT} = 18e^{-0.05/4} = 17.776$

c) ① $P = 18e^{-0.05/4} - \Delta S_0 = 2.776...$

iii) a) ③ $Z = \text{randn}(N,1); Y = \alpha \mu * \text{delta} + \beta * \text{sqrt}(\text{delta}) * Z;$
 $YC = \text{cumsum}(Y); X = X_0 + YC;$

b) ① X_N has normal distribution

c) ① $E(Z_n) = 0 \forall n \Rightarrow E(X_N) = X_0 + N\alpha\delta = X_0 + \alpha T$

② $\text{Var}(X_N) = \text{Var}(\beta\sqrt{\delta}(Z_1 + \dots + Z_N)) = \beta^2\delta [\text{Var}(Z_1) + \dots + \text{Var}(Z_N)] = \beta^2\delta N = \beta^2 T$

4) i) ① $W(R,T) = \max(0, L-R)$

ii) ① $R=0: W(0,t) = Le^{-r(T-t)}$

① $R \rightarrow \infty: W(R,t) = 0$

iii) a) ① $V_\tau = -W_t; \text{ ① } V_S = RW_R; \text{ ① } V_{SS} = RW_R + R^2 W_{RR}$

b) ② $\underbrace{W_t}_{-V_\tau} + r \underbrace{RW_R}_{V_S} + \frac{\sigma^2}{2} \underbrace{R^2 W_{RR}}_{V_{SS} - V_S} = rW$

c) ① $V(s,0) = L \max(0, 1 - e^s)$

d) ② $s \rightarrow -\infty: V(s,T) \rightarrow Le^{-rT}; s \rightarrow \infty: V(s,T) \rightarrow 0$

iv) a) ② $V_S = \frac{V_{n+1}^d - V_n^d}{\Delta s}; V_{SS} = \frac{V_{n+1}^d - 2V_n^d + V_{n-1}^d}{\Delta s^2}$

b) ① $V_\tau = \frac{V_{n+1}^d - V_n^d}{\Delta \tau}$

c) ③ A) $\alpha = \frac{\sigma^2 \Delta \tau}{2 \Delta s^2} - \frac{\Delta \tau}{2 \Delta s} (r - \frac{\sigma^2}{2}), \beta = -\Delta \tau r - \Delta \tau \frac{\sigma^2}{\Delta s^2}$
 $\gamma = \frac{\Delta \tau}{2 \Delta s} (r - \frac{\sigma^2}{2}) + \frac{\sigma^2 \Delta \tau}{2 \Delta s^2}$

c) ③ $V_n^0 = L \max(0, 1 - e^{s_n})$
 $V_n^d = Le^{-rT} V_n^0 = 0$

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Q1) i) a) $\pm 1.00, \pm 1.25, \pm 1.5, \pm 1.75, \pm 2.00,$
~~(1)~~ ~~(2)~~ (1) $\pm 2.5, \pm 3, \pm 3.5$

(1) b) 4

(1) c) 3.5

(1) d) 1

(1) ~~(2)~~ ~~(3)~~ e) $1.25 - 1 = 0.25 = \frac{1}{4}$

ii) a) For small x , the value of $\cos x$ gets rounded to 1,

hence $1 - \cos x$ is calculated as 0. The value of x can

(2) still be represented as a number $\neq 0$, hence $\frac{1 - \cos x}{x^2}$ is calculated as 0.

b) Using $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$, we obtain

(1) ~~(2)~~ ~~(3)~~
$$\frac{1 - \cos x}{x^2} = \frac{1 - \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right)}{x^2} = \frac{1}{2!} - \frac{x^2}{4!} + \frac{x^4}{6!} - \dots$$

c)
$$f(10^{-8}) = \frac{1}{2} - \frac{(10^{-8})^2}{4!} + \frac{(10^{-8})^4}{6!} - \dots \approx \frac{1}{2} - 10^{(-16)}/24 > \frac{1}{2} - \text{eps}/8$$

(2) In the double precision normalized (binary) floating point arithmetic, $1/2$ can be represented as: $(1.000\dots 0000)_2 \cdot 2^{-1}$ and the number that is just before $1/2$ can be represented as: $(1.111\dots 111)_2 \cdot 2^{-2}$. Since

$$(1.000\dots 0000)_2 \cdot 2^{-1} = 2^{-2} \cdot 2 \cdot (1.000\dots 000)_2 = 2^{-2} \cdot [(1.111\dots 111)_2 + \text{eps}],$$

$$(1.000\dots 0000)_2 \cdot 2^{-1} - (1.111\dots 111)_2 \cdot 2^{-2} = \text{eps}/4.$$

Hence, in double precision floating point arithmetic, the best approximation of $f(10^{-8})$ is $\frac{1}{2}$.

Q1) iii) a) Since C is symmetric we can store just the lower triangular part, which has

$$n + (n-1) + (n-2) + \dots + 2 + 1 = \frac{n(n+1)}{2} \text{ elements.}$$

Each number is stored using 8 bytes, hence we require

$$(2) \quad 8 \frac{n(n+1)}{2} = 4n^2 + 4n = 4 \cdot 10^6 + 4 \cdot 10^3 \text{ bytes.}$$

If the full matrix is stored we require $8n^2$ bytes,
 $\checkmark \quad 8 \cdot 10^6$ bytes.

~~15824 = 10^3~~
 b) Since $n = 10^3$, we require

$$(1) \quad \frac{(10^3)^3}{3} + C \cdot 10^6 \approx 3.33 \cdot 10^8 \text{ flops.}$$

c) The computer can do

$$\underbrace{4 \cdot 10^9}_{4 \text{ GHz}} \cdot \underbrace{4}_{\text{quadcore}} = 1 \text{ flop/sec} \Rightarrow 1.6 \cdot 10^{10} \text{ flop/sec.}$$

(2)

We require a

It takes approximately 0.02 seconds.

$$\frac{3.33 \cdot 10^8}{1.6 \cdot 10^{10}} \approx 2 \cdot 10^{-2}$$

⑤ Q2) i)

$$s_0(0) = a_0 = 2$$

$$s_1(0) = a_1 = 2$$

$$s_0'(-1) = b_0 = b_1 = s_1'(0)$$

$$s_0'(-1) = b_0 - 2c_0 = 0$$

$$s_0(-1) = a_0 - b_0 + c_0 = 1 \Rightarrow -b_0 + c_0 = -1$$

$$\Rightarrow c_0 = 1$$

$$\Rightarrow b_0 = 2$$

$$b_1 = b_0 = 2$$

$$s_1(1) = a_1 + b_1 + c_1 = 0 \Rightarrow c_1 = -4.$$

$$\text{Hence } a_0 = 2, b_0 = 2, c_0 = 1$$

$$a_1 = 2, b_1 = 2, c_1 = -4.$$

④ ii) From Lecture 06.pdf, slide 21

$$A_0 = \frac{\hat{y}_0}{n} = \frac{1}{n} \sum_{j=0}^n y_j \underbrace{e^{-2\pi i j 0/n}}_{=1} = \frac{1}{n} \sum_{j=0}^n y_j$$

$$\text{Here, } n=5, \text{ hence } A_0 = \frac{1}{5}(1+0+2+2+0) = 1.$$

$$\int_0^1 f(x) dx = \underbrace{A_0}_{=1} + \sum_{k=1}^2 \left[\underbrace{A_k \int_0^1 \cos(2\pi kx) dx}_{=0} + \underbrace{B_k \int_0^1 \sin(2\pi kx) dx}_{=0} \right] = 1.$$

Q2) iii) Let $p_1(x) = a + bx$

We have

$$\textcircled{1} \quad \int_{-1}^1 p_1(x) dx = 2a.$$

Further, for N ~~odd~~ ^{even} we have

Note: P2 should be replaced with P1

$$\begin{aligned} Q_h(p_2) &= h \left(\frac{1}{2} p_2(-1) + p_2(-1+h) + \dots + p_2(1-h) + \frac{1}{2} p_2(1) \right) \\ &= \frac{h}{2} \left(\overbrace{p_2(-1) + p_2(1)}^{2a} \right) + h \left(\overbrace{p_2(-1+h) + p_2(1-h)}^{2a} \right) \end{aligned}$$

$$\textcircled{3} \quad + \dots + h \left(\underbrace{p_2(-h) + p_2(h)}_{2a} \right) + h \underbrace{p_2(0)}_a$$

$$= \frac{h}{2} 2a + \frac{N-2}{2} 2ah + ha$$

$$= ah [1 + N - 2 + 1] = ahN = 2a.$$

For N odd we have

$$Q_h(p_2) = \frac{h}{2} \left(\underbrace{p_2(-1) + p_2(1)}_{=2a} \right) + h \left(\underbrace{p_2(-1+h) + p_2(1-h)}_{=2a} \right) + \dots + h \left(\underbrace{p_2(-h) + p_2(h)}_{=2a} \right)$$

$$\textcircled{2} \quad = \frac{h}{2} 2a + 2ah \cdot \frac{N-1}{2} = ahN = 2a.$$

Q2) (ii) b)

$$p_2(x) = a + bx + cx^2$$

$$\int_{-1}^1 p_2(x) dx = 2a + \frac{2}{3}c.$$

Consider $a = b = 0$, $c = 1$. $\int_{-1}^1 x^2 dx = \frac{2}{3}$.
For Neven, we have

$$Q_h(x^2) = h \left(\frac{1}{2}(-1)^2 + (-1+h)^2 + (-1+2h)^2 + \dots + (1-2h)^2 + (1-h)^2 + \frac{1}{2}(1)^2 \right)$$

$$= h \left(\frac{1}{2}(-1)^2 + \frac{1}{2}(1)^2 \right) + 2h \sum_{r=1}^{\frac{N-2}{2}} (1-rh)^2$$

$$= h + 2h \sum_{r=1}^{\frac{N-2}{2}} (1 - 2rh + r^2 h^2)$$

$$= h + 2h \left[\frac{N-2}{2} - 2h \frac{\left(\frac{N-2}{2}\right)\left(\frac{N-2}{2}+1\right)}{2} + h^2 \frac{\frac{N-2}{2}\left(\frac{N-2}{2}+1\right)(N-2+1)}{6} \right]$$

$$= h + hN - 2h + \frac{h^2}{2} (N-2)N + \frac{h^3}{12} (N-2)N(N-1)$$

$$= 2 - h + 2 - 2h + \frac{1}{6} \underbrace{(hN - 2h)}_{2-2h} \underbrace{(hN - h)}_{2-h}$$

$$= 4 - 3h + \frac{1}{3} (1-h)(2-h)$$

$$= 4 - 3h + \frac{1}{3} (2 - 3h + h^2)$$

$$= \frac{14}{3} - 4h + \frac{h^2}{3} \neq \frac{2}{3} \quad \text{for any } h = \frac{2}{N}, N \in \mathbb{N}.$$

Similarly for odd N .

Q3) i) a) $1 = \int_0^3 p(x) dx = c \int_0^3 x^2 dx = \frac{c}{3} x^3 \Big|_0^3 = 9c$

① $\Rightarrow c = \frac{1}{9}$

① b) $E(f(x)) = \int_0^3 f(x) p(x) dx$

c) $P(z) = \int_0^z p(x) dx = \frac{1}{9} \int_0^z x^2 dx = \frac{z^3}{27}$

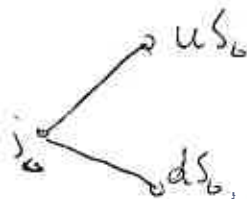
② $v = \frac{z^3}{27} \Leftrightarrow z = 3 v^{1/3}$

$\Rightarrow p^{-1}(v) = 3 v^{1/3}$

$p^{-1}: [0, 1] \rightarrow \mathbb{R}$

① d) The random variable $p^{-1}(Y) = 3 Y^{1/3}$ has distribution with prob. density fct. p .

ii) ② e) $P_u = (K - uS_0)_+$ -- price of put option at expiry when stock price goes up to uS_0
 $P_d = (K - dS_0)_+$ ① -- price of put option at expiry when stock price goes down to dS_0



Value of portfolio at expiry:

$\Delta S_0 u + (K - uS_0)_+ \quad \text{②} \quad \Delta S_0 d + (K - dS_0)_+$

$\Rightarrow \Delta = \frac{(K - dS_0)_+ - (K - uS_0)_+}{S_0(u - d)} \quad \text{③} = \frac{5 \cdot 5 - 0}{11} = \frac{1}{2}$

Q3) ii)

b) Value of portfolio at time T:

$$\Delta S_0 d + (K - dS_0)_+ = \frac{1}{2} 30 \frac{5}{6} + \left(30.5 - \frac{5}{6} 30\right)_+$$

$$(2) \quad = \frac{25 + 11}{2} = 18$$

$$\text{Value of portfolio at time 0: } 18 \cdot e^{-rT} = 18 \cdot e^{-0.05/4}$$

c) Option price at time 0: P

$$\Delta S_0 + P = 18 \cdot e^{-0.05/4}$$

$$(1) \quad \Rightarrow P = 18 \cdot e^{-0.05/4} - 15 \approx 2.776 \dots$$

Q3) iii) a)

$$\text{We have } X_1 = X_0 + \alpha \delta + \beta \sqrt{\delta} Z_1$$

$$X_2 = X_0 + 2\alpha \delta + \beta \sqrt{\delta} (Z_1 + Z_2)$$

$$X_3 = X_0 + 3\alpha \delta + \beta \sqrt{\delta} (Z_1 + Z_2 + Z_3)$$

$$\vdots$$

$$X_n = X_0 + n\alpha \delta + \beta \sqrt{\delta} (Z_1 + \dots + Z_n)$$

Code: $N = \dots;$

$\alpha = \dots;$

$\beta = \dots;$

$X_0 = \dots;$

$T = \dots;$

$\delta = T/N.$

$$\textcircled{3} \quad Z = \text{randn}(N, 1);$$

$$Y = \alpha * \delta + \beta * \text{sqrt}(\delta) * Z;$$

$$Y_c = \text{cumsum}(Y);$$

$$X = X_0 + Y_c;$$

① b) X_N has a normal distribution as it is a sum of normal random variables plus some constants;

c) $E(Z_n) = 0 \quad \forall n$; Hence

$$\textcircled{1} \quad E(X_N) = X_0 + N\alpha \delta = X_0 + \alpha T$$

$$\textcircled{2} \quad \text{Var}(X_N) = \text{Var}(\beta \sqrt{\delta} (Z_1 + \dots + Z_N)) = \beta^2 \delta [\text{Var}(Z_1) + \dots + \text{Var}(Z_N)]$$

$$= \beta^2 \delta N = \beta^2 T.$$

(Q4) i) 'Initial' conditions

① At expiry time T : $W(R, T) = \max(0, L - R)$

ii) $R=0$: $W(0, t) = L e^{-r(T-t)}$

② $R \rightarrow \infty$: $W(R, t) = 0$

iii) a) $\frac{\partial V}{\partial \tau} = \frac{\partial}{\partial \tau} (W(R, T-\tau)) = - \frac{\partial W}{\partial t} (R, T-\tau)$

③

$$\frac{\partial V}{\partial s} = \frac{\partial}{\partial s} (W(L e^s, T-\tau)) = \frac{\partial W}{\partial R} \frac{\partial (L e^s)}{\partial s} = R \frac{\partial W}{\partial R}$$

$$\frac{\partial^2 V}{\partial s^2} = \frac{\partial}{\partial s} \left(R \frac{\partial W}{\partial R} \right) = R \frac{\partial W}{\partial R} + R \frac{\partial}{\partial s} \left(\frac{\partial W}{\partial R} \right) =$$

$$= R \frac{\partial W}{\partial R} + R \frac{\partial}{\partial R} \left(\frac{\partial W}{\partial R} \right) \frac{\partial R}{\partial s} = R \frac{\partial W}{\partial R} + R^2 \frac{\partial^2 W}{\partial R^2}$$

$$\Rightarrow R^2 \frac{\partial^2 W}{\partial R^2} = \frac{\partial^2 V}{\partial s^2} - R \frac{\partial W}{\partial R} = \frac{\partial^2 V}{\partial s^2} - \frac{\partial V}{\partial s}$$

Q4) i)

b)

$$\underbrace{\frac{\partial W}{\partial t}}_{-\frac{\partial V}{\partial \tau}} + r \underbrace{R \frac{\partial W}{\partial R}}_{\frac{\partial V}{\partial s}} + \frac{\sigma^2}{2} \underbrace{R^2 \frac{\partial^2 W}{\partial R^2}}_{\frac{\partial^2 V}{\partial s^2} - \frac{\partial V}{\partial s}} = r W$$

$$\textcircled{2} \Rightarrow -\frac{\partial V}{\partial \tau} + r \frac{\partial V}{\partial s} + \frac{\sigma^2}{2} \left(\frac{\partial^2 V}{\partial s^2} - \frac{\partial V}{\partial s} \right) = r V$$

$$\Rightarrow -\frac{\partial V}{\partial \tau} + \left(r - \frac{\sigma^2}{2} \right) \frac{\partial V}{\partial s} + \frac{\sigma^2}{2} \frac{\partial^2 V}{\partial s^2} = r V$$

$$\textcircled{1} \text{ c) } V(s, 0) = \max(0, L - L e^s) = L \max(0, 1 - e^s)$$

$$\text{d) } s \rightarrow -\infty : V(s, \tau) \xrightarrow{s \rightarrow -\infty} L e^{-r\tau}$$

$$\textcircled{2} \quad s \rightarrow \infty : V(s, \tau) \xrightarrow{s \rightarrow \infty} 0$$

Q4) iv)

$$\textcircled{1} \text{ a) } \frac{\partial V}{\partial s}(s_n, \tau_j) \approx \frac{V_{n+1}^j - V_{n-1}^j}{2\Delta s}$$

$$\textcircled{1} \frac{\partial^2 V}{\partial s^2}(s_n, \tau_j) \approx \frac{V_{n+1}^j - 2V_n^j + V_{n-1}^j}{\Delta s^2}$$

$$\textcircled{1} \text{ b) } \frac{\partial V}{\partial \tau}(s_n, \tau_j) \approx \frac{V_n^{j+1} - V_n^j}{\Delta \tau}$$

c) A)

$$- \frac{V_n^{j+1} - V_n^j}{\Delta \tau} + \left(r - \frac{\sigma^2}{2}\right) \frac{V_{n+1}^j - V_{n-1}^j}{2\Delta s} + \frac{\sigma^2}{2} \frac{V_{n+1}^j - 2V_n^j + V_{n-1}^j}{\Delta s^2}$$

$$= r V_n^j$$

$$V_n^{j+1} = V_n^j - \Delta \tau r V_n^j - \Delta \tau \frac{\sigma^2}{2\Delta s^2} V_n^j$$

$$+ \left(r - \frac{\sigma^2}{2}\right) \frac{\Delta \tau}{2\Delta s} V_{n+1}^j + \frac{\sigma^2 \Delta \tau}{2\Delta s^2} V_{n+1}^j$$

$$- \left(r - \frac{\sigma^2}{2}\right) \frac{\Delta \tau}{2\Delta s} V_{n-1}^j + \frac{\sigma^2 \Delta \tau}{2\Delta s^2} V_{n-1}^j$$

Q4) iv) c) A)

Thus

$$\begin{aligned}
 V_n^{j+1} &= V_{n-1}^j \left[\frac{\sigma^2 \Delta \tau}{2 \Delta s^2} - \frac{\Delta \tau}{2 \Delta s} \left(r - \frac{\sigma^2}{2} \right) \right] \\
 &\quad + V_n^j \left[1 - \Delta \tau r - \Delta \tau \frac{\sigma^2}{\Delta s^2} \right] \\
 &\quad + V_{n+1}^j \left[\frac{\Delta \tau}{2 \Delta s} \left(r - \frac{\sigma^2}{2} \right) + \frac{\sigma^2 \Delta \tau}{2 \Delta s^2} \right]
 \end{aligned}$$

$$B) V_n^0 = L \max(0, 1 - e^{s_n})$$

$$V_0^j = L e^{-r \tau_j}$$

$$V_{M+1}^j = 0$$